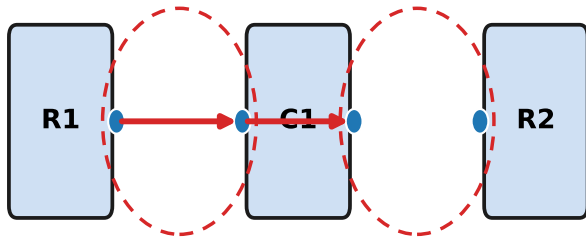
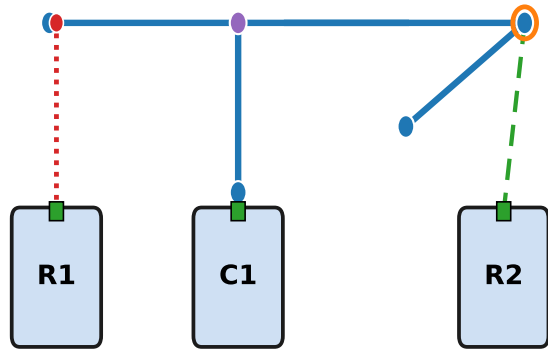


(a) Radius-based joining (production) (b) Endpoint-graph joining (ours): five typed edges



*radius grabs every pin in range  
→ R1, C1, R2 wrongly merged*



- |                                                                                               |                                                                                                               |                                                                                                         |                                                                                                   |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|  Wire body    |  Endpoint-endpoint           |  Rail-tap (pin-body) |  Component pin |
|  Endpoint-pin |  T-junction (endpoint-body) |  Wire endpoint       |                                                                                                   |